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1 Atoms by type

Atomic proportions not counting hydrogen

Note that QM9 is the only base with more O than N

	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
C	71.89	69.43	67.26	69.02	71.84
N	11.87	14.42	16.71	15.68	15.56
O	15.96	14.29	11.83	11.38	11.87
F	0.28	1.85	4.2	3.92	0.00
S	0.00	0.00	0.00	0.00	0.65
Cl	0.00	0.00	0.00	0.00	0.07

Hydrogen proportion

	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
H	51.09	54.77	51.78	51.16	53.83

2 Atoms by hybridization

..pending..one table for C, another for N..

3 Bonds by type

Hol' up: any J−H bond can be mutated to J−F. Is this a special kind of insertion? By that same token, any J−H can be changed to a J≡N or J=O depending on how many avaliable bonds J has. This is interesting! Sounds like a reasonable approach so long as this is done with low probability. Maybe draw these probabilities from the proportion of C/N/O/F, and this will lead to largely insertions of C. I think this warrants doing ANOTHER scan now checking the amount of C of every hybridization, and so on? Then using their proportions as 'probability points'

Looks like PC9 is the only dataset with bonds between a heteroatom and F. The other datasets seem to only do C−F

C−F	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.26	1.83	4.22	3.78	0.00

O−F	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.02	0.00	0.00	0.00

N−F	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.07	0.00	0.00	0.00

F−F	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	1/774611	1/4595114	1/297057239	0.00

O−O	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.22	0.00	0.00	0.00
d	0.00	1/774611	1/4595114	1/297057239	0.00
a	0.00	21/774611	0.00	0.00	0.00

N−O	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.16	1.35	1.57	1.39	1.19
d	0.01	0.19	0.01	0.01	0.01
a	0.40	0.17	0.32	0.19	0.17

N−N	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	1.28	1.81	1.66	1.43
d	0.00	0.06	0.01	0.01	103/2842418
t	0.00	1/774611	0.00	0.00	0.00
a	0.88	0.75	0.85	0.56	0.33

Unsurprisingly, QM9 has the largest C−O proportion out of all the bases, and also the smallest C−N proportion

C−O	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	14.96	13.26	10.68	10.76	11.11
d	4.45	3.78	3.42	2.94	3.11
a	1.21	0.64	0.8	0.53	0.59

C−N	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	11.07	15.96	14.87	15.05	16.13
d	1.65	1.87	4.22	3.89	3.64
t	1.39	0.75	0.79	0.63	0.8
a	4.44	3.61	3.32	2.47	2.46

C−C	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	52.66	44.64	43.67	47.3	46.91
d	1.4	4.94	5.88	5.88	6.09
t	1.55	0.57	1.31	1.08	1.59
a	3.49	4.02	2.24	1.84	2.7

Only GDB-13 has Cl and S

C−Cl	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.00	0.00	0.00	0.07

O−S	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.00	0.00	0.00	0.04
d	0.00	0.00	0.00	0.00	0.46

N−S	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.00	0.00	0.00	0.12
a	0.00	0.00	0.00	0.00	0.15

C−S	Percentage				
Type	QM9	PC9	GDB-11$_{<9}$	GDB-11	GDB-13$_{<9}$
s	0.00	0.00	0.00	0.00	0.31
d	0.00	0.00	0.00	0.00	0.08
a	0.00	0.00	0.00	0.00	0.51

4 Full-dataset tables (atoms)

4.1 QM9

				Percentage	
i	Atom	Cardinality		heavy	all
	Σ_{h}	1.18E6	1177631	100	—
	$\Sigma_{\forall}$	2.41E6	2407753	—	100
1	C	8.47E5	846556	71.89	35.16
2	O	1.88E5	187997	15.96	7.81
3	N	1.40E5	139764	11.87	5.8
4	F	3.31E3	3314	0.28	0.14
5	H	1.23E6	1230122	0.00	51.09

4.2 PC9

				Percentage	
i	Atom	Cardinality		heavy	all
	Σ_{h}	8.04E5	803528	100	—
	$\Sigma_{\forall}$	1.78E6	1776364	—	100
1	C	5.58E5	557927	69.43	31.41
2	N	1.16E5	115865	14.42	6.52
3	O	1.15E5	114848	14.29	6.47
4	F	1.49E4	14888	1.85	0.84
5	H	9.73E5	972836	0.00	54.77

4.3 GDB-11

size 1 to 9

				Percentage	
i	Atom	Cardinality		heavy	all
	Σ_{h}	4.62E6	4619675	100	—
	$\Sigma_{\forall}$	9.58E6	9581191	—	100
1	C	3.11E6	3107006	67.26	32.43
2	N	7.72E5	772109	16.71	8.06
3	O	5.47E5	546663	11.83	5.71
4	F	1.94E5	193897	4.2	2.02
5	H	4.96E6	4961516	0.00	51.78

size 1 to 11

				Percentage	
i	Atom	Cardinality		heavy	all
	Σ_{h}	2.87E8	286520241	100	—
	$\Sigma_{\forall}$	5.87E8	586650981	—	100
1	C	1.98E8	197753383	69.02	33.71
2	N	4.49E7	44923614	15.68	7.66
3	O	3.26E7	32611754	11.38	5.56
4	F	1.12E7	11231490	3.92	1.91
5	H	3.00E8	300130740	0.00	51.16

4.4 GDB-13

size 1 to 9

i	Atom	Cardinality		Percentage heavy	Percentage all
	Σ_{h}	2.82E6	2819104	100	—
	$\Sigma_{\forall}$	6.11E6	6105833	—	100
1	C	2.03E6	2025329	71.84	33.17
2	N	4.39E5	438777	15.56	7.19
3	O	3.35E5	334700	11.87	5.48
4	S	1.83E4	18292	0.65	0.3
5	Cl	2.01E3	2006	0.07	0.03
6	H	3.29E6	3286729	0.00	53.83

5 Full-dataset tables (hat-hat bonds)

5.1 QM9

i	Type	Cardinality		Percentage
	Σ	1.26E6	1258966	100
1	C C s	6.63E5	663029	52.66
2	C O s	1.88E5	188373	14.96
3	C N s	1.39E5	139309	11.07
4	C O d	5.60E4	56040	4.45
5	C N a	5.59E4	55863	4.44
6	C C a	4.40E4	43953	3.49
7	C N d	2.08E4	20799	1.65
8	C C t	1.95E4	19527	1.55
9	C C d	1.76E4	17616	1.4
10	C N t	1.75E4	17500	1.39
11	C O a	1.53E4	15294	1.21
12	N N a	1.11E4	11085	0.88
13	N O a	5.05E3	5054	0.4
14	C F s	3.31E3	3314	0.26
15	N O s	2.07E3	2068	0.16
16	N O d	1.42E2	142	0.01

5.2 PC9

i	Type	Cardinality		Percentage
	Σ	7.75E5	774611	100
1	C C s	3.46E5	345798	44.64
2	C N s	1.24E5	123594	15.96
3	C O s	1.03E5	102707	13.26
4	C C d	3.83E4	38298	4.94
5	C C a	3.12E4	31170	4.02
6	C O d	2.93E4	29302	3.78
7	C N a	2.80E4	27989	3.61
8	C N d	1.45E4	14518	1.87
9	C F s	1.42E4	14178	1.83
10	N O s	1.05E4	10459	1.35
11	N N s	9.92E3	9921	1.28
12	N N a	5.83E3	5825	0.75
13	C N t	5.77E3	5774	0.75
14	C O a	4.98E3	4978	0.64
15	C C t	4.41E3	4408	0.57
16	O O s	1.72E3	1720	0.22
17	N O d	1.46E3	1460	0.19
18	N O a	1.31E3	1308	0.17
19	N F s	5.14E2	514	0.07
20	N N d	4.73E2	473	0.06
21	O F s	1.93E2	193	0.02
22	O O a	2.10E1	21	0.0
23	N N t	1.00	1	0.0
24	O O d	1.00	1	0.0
25	F F s	1.00	1	0.0

5.3 GDB-11

size 1 to 9

i	Type	Cardinality		Percentage
	Σ	4.60E6	4595114	100
1	C C s	2.01E6	2006471	43.67
2	C N s	6.83E5	683405	14.87
3	C O s	4.91E5	490650	10.68
4	C C d	2.70E5	270384	5.88
5	C N d	1.94E5	194067	4.22
6	C F s	1.94E5	193894	4.22
7	C O d	1.57E5	157001	3.42
8	C N a	1.53E5	152649	3.32
9	C C a	1.03E5	102964	2.24
10	N N s	8.34E4	83399	1.81
11	N O s	7.20E4	72022	1.57
12	C C t	6.04E4	60351	1.31
13	N N a	3.89E4	38875	0.85
14	C O a	3.69E4	36888	0.8
15	C N t	3.63E4	36258	0.79
16	N O a	1.48E4	14774	0.32
17	N N d	5.33E2	533	0.01
18	N O d	5.27E2	527	0.01
19	F F s	1.00	1	0.0
20	O O d	1.00	1	0.0

size 1 to 11

i	Type	Cardinality		Percentage
	Σ	2.97E8	297057239	100
1	C C s	1.41E8	140520094	47.3
2	C N s	4.47E7	44712429	15.05
3	C O s	3.20E7	31974592	10.76
4	C C d	1.75E7	17475496	5.88
5	C N d	1.16E7	11553389	3.89
6	C F s	1.12E7	11231487	3.78
7	C O d	8.75E6	8747522	2.94
8	C N a	7.33E6	7331958	2.47
9	C C a	5.47E6	5474385	1.84
10	N N s	4.93E6	4928734	1.66
11	N O s	4.13E6	4132396	1.39
12	C C t	3.21E6	3211076	1.08
13	C N t	1.87E6	1873197	0.63
14	N N a	1.67E6	1673414	0.56
15	C O a	1.58E6	1581474	0.53
16	N O a	5.69E5	569192	0.19
17	N N d	3.58E4	35784	0.01
18	N O d	3.06E4	30618	0.01
19	F F s	1.00	1	0.0
20	O O d	1.00	1	0.0

5.4 GDB-13

size 1 to 9

i	Type	Cardinality		Percentage
	Σ	2.84E6	2842418	100
1	C C s	1.33E6	1333241	46.91
2	C N s	4.59E5	458603	16.13
3	C O s	3.16E5	315798	11.11
4	C C d	1.73E5	173046	6.09
5	C N d	1.03E5	103481	3.64
6	C O d	8.83E4	88272	3.11
7	C C a	7.67E4	76731	2.7
8	C N a	6.98E4	69826	2.46
9	C C t	4.51E4	45083	1.59
10	N N s	4.08E4	40775	1.43
11	N O s	3.39E4	33921	1.19
12	C N t	2.28E4	22834	0.8
13	C O a	1.67E4	16727	0.59
14	C S a	1.44E4	14403	0.51
15	O S d	1.31E4	13124	0.46
16	N N a	9.38E3	9380	0.33
17	C S s	8.79E3	8792	0.31
18	N O a	4.71E3	4705	0.17
19	N S a	4.36E3	4357	0.15
20	N S s	3.42E3	3418	0.12
21	C S d	2.24E3	2242	0.08
22	C Cl s	2.01E3	2006	0.07
23	O S s	1.13E3	1130	0.04
24	N O d	4.20E2	420	0.01
25	N N d	1.03E2	103	0.0

6 Full-dataset tables (light-atom bonds)

6.1 QM9

i	Type	Cardinality		Percentage
	Σ	1.23E6	1230122	100
1	H C s	1.10E6	1104294	89.77
2	H N s	7.45E4	74514	6.06
3	H O s	5.13E4	51314	4.17

6.2 PC9

i	Type	Cardinality		Percentage
	Σ	9.73E5	972836	100
1	H C s	8.54E5	853847	87.77
2	H N s	7.70E4	77024	7.92
3	H O s	4.20E4	41964	4.31
4	H F s	1.00	1	0.0

6.3 GDB-11

size 1 to 9

i	Type	Cardinality		Percentage
	Σ	4.96E6	4961516	100
1	H C s	4.22E6	4221804	85.09
2	H N s	5.76E5	576096	11.61
3	H O s	1.64E5	163615	3.3
4	H F s	1.00	1	0.0

size 1 to 11

i	Type	Cardinality		Percentage
	Σ	3.00E8	300130740	100
1	H C s	2.58E8	257994516	85.96
2	H N s	3.27E7	32746665	10.91
3	H O s	9.39E6	9389558	3.13
4	H F s	1.00	1	0.0

6.4 GDB-13

size 1 to 9

i	Type	Cardinality		Percentage
	Σ	3.29E6	3286729	100
1	H C s	2.86E6	2857167	86.93
2	H N s	3.36E5	336409	10.24
3	H O s	9.32E4	93153	2.83

7 Raw `bondcow` output (atoms)

7.1 QM9

```
smis1-qm9.smi
1177631 1230122 2407753
0 0.0

== atom counts
5
1 ; C ;  846556 71.89 35.16
2 ; O ;  187997 15.96  7.81
3 ; N ;  139764 11.87   5.8
4 ; F ;    3314  0.28  0.14
5 ; H ; 1230122  0.00 51.09
```

7.2 PC9

```
smis1-pc9.smi
803528 972836 1776364
1234 1.24

== atom counts
5
1 ; C ; 557927 69.43 31.41
2 ; N ; 115865 14.42  6.52
3 ; O ; 114848 14.29  6.47
4 ; F ;  14888  1.85  0.84
```

```
5 ; H ; 972836  0.00 54.77
```

7.3 GDB-11

size 1 to 9

```
gdb11_size1to9.smi
4619675 4961516 9581191
0 0.0

== atom counts
5
1 ; C ; 3107006 67.26 32.43
2 ; N ;  772109 16.71  8.06
3 ; O ;  546663 11.83  5.71
4 ; F ;  193897   4.2  2.02
5 ; H ; 4961516  0.00 51.78
```

size 10 to 11

```
gdb11_size10to11.smi
281900566 295169224 577069790
0 0.0

== atom counts
5
1 ; C ; 194646377 69.05 33.73
2 ; N ;  44151505 15.66  7.65
3 ; O ;  32065091 11.37  5.56
4 ; F ;  11037593  3.92  1.91
5 ; H ; 295169224  0.00 51.15
```

size 1 to 11

```
gdb11_size1to11.smi
286520241 300130740 586650981
0 0.0

== atom counts
5
1 ; C ; 197753383 69.02 33.71
2 ; N ;  44923614 15.68  7.66
3 ; O ;  32611754 11.38  5.56
4 ; F ;  11231490  3.92  1.91
5 ; H ; 300130740  0.00 51.16
```

7.4 GDB-13

size 1 to 9

```
gdb13_size1to9.smi
2819104 3286729 6105833
0 0.0
```

```
== atom counts
6
1 ; C  ; 2025329 71.84 33.17
2 ; N  ;  438777 15.56  7.19
3 ; O  ;  334700 11.87  5.48
4 ; S  ;   18292  0.65   0.3
5 ; Cl ;    2006  0.07  0.03
6 ; H  ; 3286729  0.00 53.83
```

8 Raw bondcow output (bonds)

8.1 QM9

```
smis1-qm9.smi
1258966 1230122 2489088
0 0.0

== hat-hat bonds
16
 1 ; C C s ; 663029 52.66 26.64
 2 ; C O s ; 188373 14.96  7.57
 3 ; C N s ; 139309 11.07   5.6
 4 ; C O d ;  56040  4.45  2.25
 5 ; C N a ;  55863  4.44  2.24
 6 ; C C a ;  43953  3.49  1.77
 7 ; C N d ;  20799  1.65  0.84
 8 ; C C t ;  19527  1.55  0.78
 9 ; C C d ;  17616   1.4  0.71
10 ; C N t ;  17500  1.39   0.7
11 ; C O a ;  15294  1.21  0.61
12 ; N N a ;  11085  0.88  0.45
13 ; N O a ;   5054   0.4   0.2
14 ; C F s ;   3314  0.26  0.13
15 ; N O s ;   2068  0.16  0.08
16 ; N O d ;    142  0.01  0.01

== light-atom bonds
3
 1 ; H C s ; 1104294 89.77 44.37
 2 ; H N s ;   74514  6.06  2.99
 3 ; H O s ;   51314  4.17  2.06

== Anomalies
0
```

8.2 PC9

```
smis1-pc9.smi
774611 972836 1747447
1234 1.24

== hat-hat bonds
25
```

```
 1 ; C C s ; 345798 44.64 19.79
 2 ; C N s ; 123594 15.96  7.07
 3 ; C O s ; 102707 13.26  5.88
 4 ; C C d ;  38298  4.94  2.19
 5 ; C C a ;  31170  4.02  1.78
 6 ; C O d ;  29302  3.78  1.68
 7 ; C N a ;  27989  3.61   1.6
 8 ; C N d ;  14518  1.87  0.83
 9 ; C F s ;  14178  1.83  0.81
10 ; N O s ;  10459  1.35   0.6
11 ; N N s ;   9921  1.28  0.57
12 ; N N a ;   5825  0.75  0.33
13 ; C N t ;   5774  0.75  0.33
14 ; C O a ;   4978  0.64  0.28
15 ; C C t ;   4408  0.57  0.25
16 ; O O s ;   1720  0.22   0.1
17 ; N O d ;   1460  0.19  0.08
18 ; N O a ;   1308  0.17  0.07
19 ; N F s ;    514  0.07  0.03
20 ; N N d ;    473  0.06  0.03
21 ; O F s ;    193  0.02  0.01
22 ; O O a ;     21   0.0   0.0
23 ; N N t ;      1   0.0   0.0
24 ; O O d ;      1   0.0   0.0
25 ; F F s ;      1   0.0   0.0

== light-atom bonds
4
 1 ; H C s ; 853847 87.77 48.86
 2 ; H N s ;  77024  7.92  4.41
 3 ; H O s ;  41964  4.31   2.4
 4 ; H F s ;      1   0.0   0.0

== Anomalies
0
```

8.3 GDB-11

```
gdb11_size1to9.smi
4595114 4961516 9556630
0 0.0

== hat-hat bonds
20
 1 ; C C s ; 2006471 43.67  21.0
 2 ; C N s ;  683405 14.87  7.15
 3 ; C O s ;  490650 10.68  5.13
 4 ; C C d ;  270384  5.88  2.83
 5 ; C N d ;  194067  4.22  2.03
 6 ; C F s ;  193894  4.22  2.03
 7 ; C O d ;  157001  3.42  1.64
 8 ; C N a ;  152649  3.32   1.6
 9 ; C C a ;  102964  2.24  1.08
```

```
10 ; N N s ;   83399  1.81  0.87
11 ; N O s ;   72022  1.57  0.75
12 ; C C t ;   60351  1.31  0.63
13 ; N N a ;   38875  0.85  0.41
14 ; C O a ;   36888   0.8  0.39
15 ; C N t ;   36258  0.79  0.38
16 ; N O a ;   14774  0.32  0.15
17 ; N N d ;     533  0.01  0.01
18 ; N O d ;     527  0.01  0.01
19 ; F F s ;       1   0.0   0.0
20 ; O O d ;       1   0.0   0.0

== light-atom bonds
4
 1 ; H C s ; 4221804 85.09 44.18
 2 ; H N s ;  576096 11.61  6.03
 3 ; H O s ;  163615   3.3  1.71
 4 ; H F s ;       1   0.0   0.0

== Anomalies
0

gdb11_size10to11.smi
292462125 295169224 587631349
0 0.0

== hat-hat bonds
18
 1 ; C C s ; 138513623 47.36 23.57
 2 ; C N s ;  44029024 15.05  7.49
 3 ; C O s ;  31483942 10.77  5.36
 4 ; C C d ;  17205112  5.88  2.93
 5 ; C N d ;  11359322  3.88  1.93
 6 ; C F s ;  11037593  3.77  1.88
 7 ; C O d ;   8590521  2.94  1.46
 8 ; C N a ;   7179309  2.45  1.22
 9 ; C C a ;   5371421  1.84  0.91
10 ; N N s ;   4845335  1.66  0.82
11 ; N O s ;   4060374  1.39  0.69
12 ; C C t ;   3150725  1.08  0.54
13 ; C N t ;   1836939  0.63  0.31
14 ; N N a ;   1634539  0.56  0.28
15 ; C O a ;   1544586  0.53  0.26
16 ; N O a ;    554418  0.19  0.09
17 ; N N d ;     35251  0.01  0.01
18 ; N O d ;     30091  0.01  0.01

== light-atom bonds
3
 1 ; H C s ; 253772712 85.98 43.19
 2 ; H N s ;  32170569  10.9  5.47
 3 ; H O s ;   9225943  3.13  1.57
```

```
== Anomalies
0

gdb11_size1to11.smi
297057239 300130740 597187979
0 0.0

== hat-hat bonds
20
 1 ; C C s ; 140520094  47.3 23.53
 2 ; C N s ;  44712429 15.05  7.49
 3 ; C O s ;  31974592 10.76  5.35
 4 ; C C d ;  17475496  5.88  2.93
 5 ; C N d ;  11553389  3.89  1.93
 6 ; C F s ;  11231487  3.78  1.88
 7 ; C O d ;   8747522  2.94  1.46
 8 ; C N a ;   7331958  2.47  1.23
 9 ; C C a ;   5474385  1.84  0.92
10 ; N N s ;   4928734  1.66  0.83
11 ; N O s ;   4132396  1.39  0.69
12 ; C C t ;   3211076  1.08  0.54
13 ; C N t ;   1873197  0.63  0.31
14 ; N N a ;   1673414  0.56  0.28
15 ; C O a ;   1581474  0.53  0.26
16 ; N O a ;    569192  0.19   0.1
17 ; N N d ;     35784  0.01  0.01
18 ; N O d ;     30618  0.01  0.01
19 ; F F s ;         1   0.0   0.0
20 ; O O d ;         1   0.0   0.0

== light-atom bonds
4
 1 ; H C s ; 257994516 85.96  43.2
 2 ; H N s ;  32746665 10.91  5.48
 3 ; H O s ;   9389558  3.13  1.57
 4 ; H F s ;         1   0.0   0.0

== Anomalies
0
```

8.4 GDB-13

```
gdb13_size1to9.smi
2842418 3286729 6129147
0 0.0

== hat-hat bonds
25
 1 ; C C s ; 1333241 46.91 21.75
 2 ; C N s ;  458603 16.13  7.48
 3 ; C O s ;  315798 11.11  5.15
 4 ; C C d ;  173046  6.09  2.82
 5 ; C N d ;  103481  3.64  1.69
```

```
 6 ; C O d ;   88272  3.11  1.44
 7 ; C C a ;   76731   2.7  1.25
 8 ; C N a ;   69826  2.46  1.14
 9 ; C C t ;   45083  1.59  0.74
10 ; N N s ;   40775  1.43  0.67
11 ; N O s ;   33921  1.19  0.55
12 ; C N t ;   22834   0.8  0.37
13 ; C O a ;   16727  0.59  0.27
14 ; C S a ;   14403  0.51  0.23
15 ; O S d ;   13124  0.46  0.21
16 ; N N a ;    9380  0.33  0.15
17 ; C S s ;    8792  0.31  0.14
18 ; N O a ;    4705  0.17  0.08
19 ; N S a ;    4357  0.15  0.07
20 ; N S s ;    3418  0.12  0.06
21 ; C S d ;    2242  0.08  0.04
22 ; C Cl s;    2006  0.07  0.03
23 ; O S s ;    1130  0.04  0.02
24 ; N O d ;     420  0.01  0.01
25 ; N N d ;     103   0.0   0.0

== light-atom bonds
3
 1 ; H C s ; 2857167 86.93 46.62
 2 ; H N s ;  336409 10.24  5.49
 3 ; H O s ;   93153  2.83  1.52

== Anomalies
0
```